Dynamics 3 Laboratory Reportand Eigenmode extraction using Abaqus CAE and experimental comparison

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1 Introduction:

Structural vibrations are an important consideration in aerospace engineering, as they can affect the durability, noise levels, and overall stability of aircraft structures. Understanding how a structure vibrates helps in predict and prevent potential problems such as resonance and fatigue failure. In this laboratory exercise, the natural frequencies and mode shapes eigenmodes of a simplified aircraft structure are studied using both experimental testing and numerical simulation. First, the eigenfrequencies and corresponding mode shapes are identified experimentally using a shaker and accelerometer setup. The same structure is then modelled in **Abaqus CAE**, where an eigenfrequency analysis is performed using the finite element method. Finally, the experimental and numerical results are compared to evaluate how well the simulation represents the real structural behaviour and to highlight any limitations of the numerical model.

2 Experimental Method:

The experimental investigation was carried out on a scaled aircraft structure using a shaker and an accelerometer. The shaker was permanently attached at station 11 and was used to apply harmonic excitation over a range of frequencies. The input force was measured using a force transducer, while the structural response was measured using an accelerometer. To identify the eigenfrequencies, the accelerometer was initially placed at station 1 and the excitation frequency was varied. Resonance was identified by observing large response amplitudes in the accelerometer signal, indicating that the excitation frequency was close to a natural frequency of the structure. Once an eigenfrequency was identified, the excitation frequency was fixed and the accelerometer was moved sequentially between stations 1 and 11. At each station, the response amplitude and phase relative to the shaker signal were recorded. Responses that were in phase with the shaker were assigned a positive sign, while out of phase responses were assigned a negative sign. The measured voltage amplitude was taken as proportional to the relative modal displacement at each station, allowing the experimental mode shapes to be reconstructed.

3 Experimental Eigenmode Reconstruction:

The measured amplitudes and phase information are used to reconstruct experimental mode shapes. Each mode shape is represented as a discrete plot of amplitude versus station number.

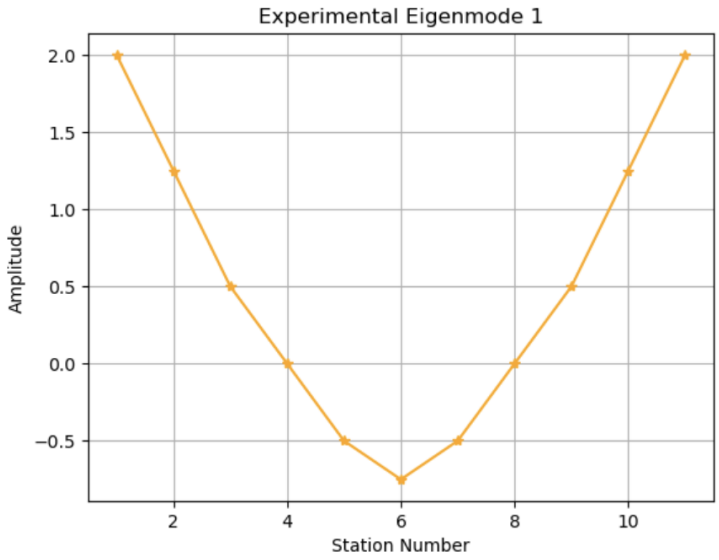


Figure 1: Experimental eigenmode 1: reconstructed from measured amplitudes and phase information across stations 1 to 11.

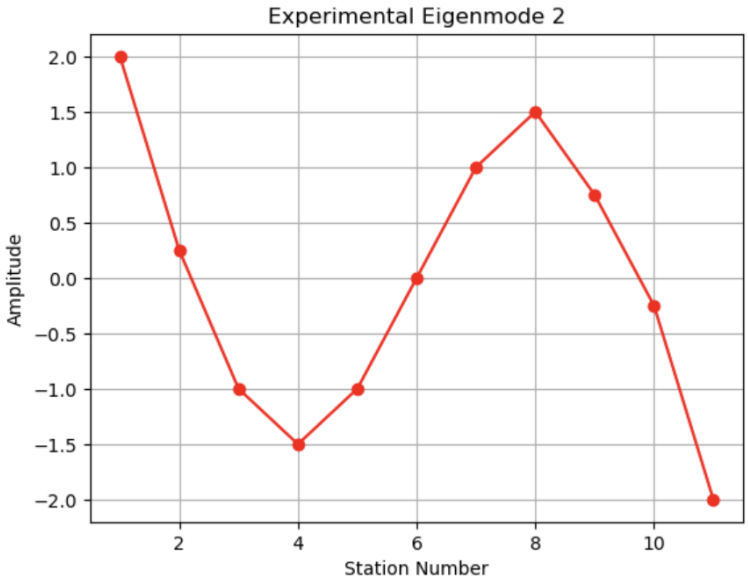


Figure 2: Experimental eigenmode 2: showing higher order bending behaviour and phase reversal across the structure.

These plots show the overall shape of the eigenmodes rather than the exact displacement values. Small asymmetries and irregularities are expected and are mainly due to experimental noise and the influence of neighbouring modes.

4 Finite Element Model:

A finite element model of the aircraft structure was created using Abaqus CAE. The model geometry represents the lab experimental structure and includes the fuselage, wings, and tail. Solid elements were used where appropriate to model the different structural components. The material was assumed to be linear elastic and isotropic. The model was set up to represent the experimental conditions, and an eigenfrequency analysis was performed to obtain the natural frequencies and corresponding mode shapes of the structures.

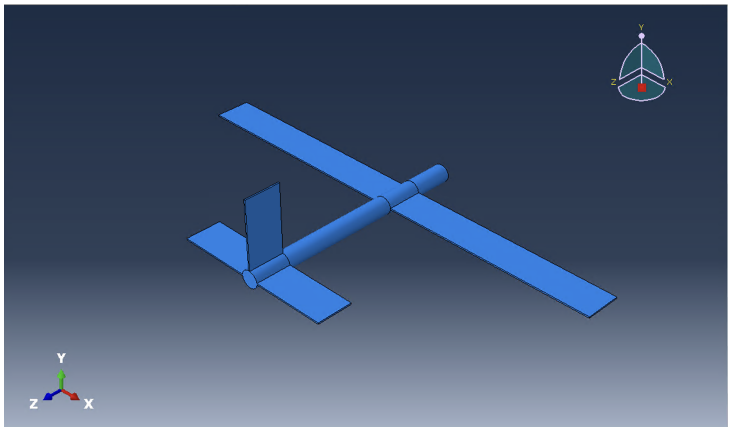


Figure 3: Finite element model of the aircraft structure in Abaqus CAE showing assembled geometry and its mesh.

5 Numerical Eigenmode Results:

The first five eigenmodes obtained from Abaqus are shown below, illustrating the characteristic vibration shapes of the structure at different sets of frequencies.

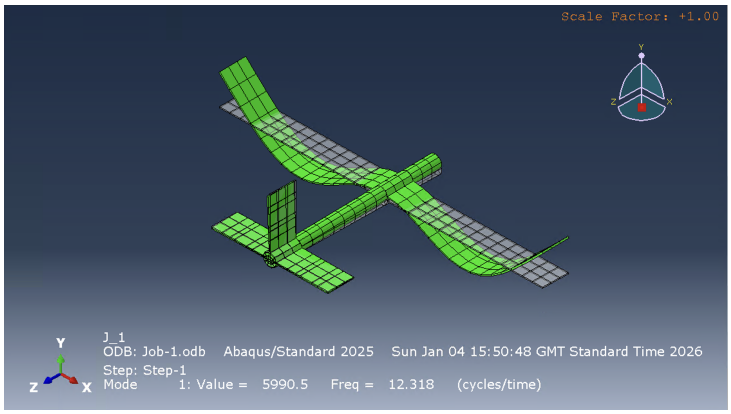


Figure 4: Abaqus eigenmode 1: at 12.3 Hz showing global bending of the structure.

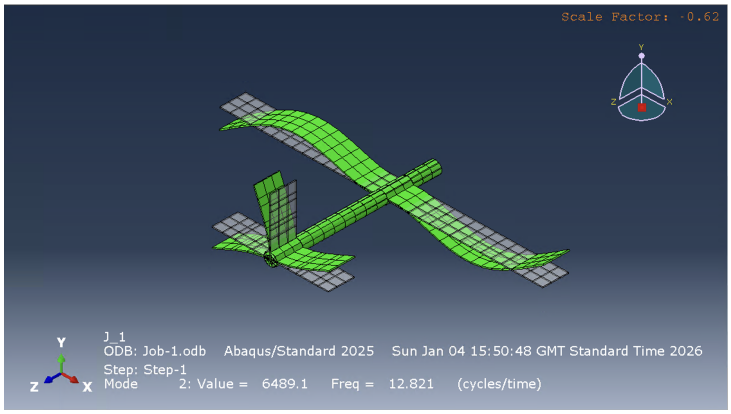


Figure 5: Abaqus eigenmode 2: at 12.8 Hz showing asymmetric bending behaviour.

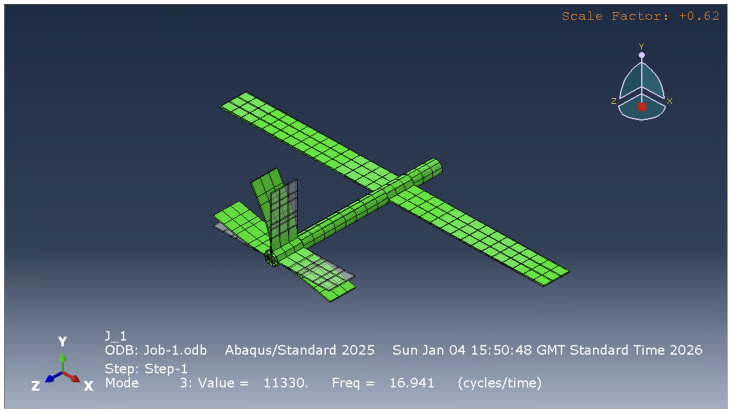


Figure 6: Abaqus eigenmode 3: at 16.9 Hz representing a higher-order deformation pattern.

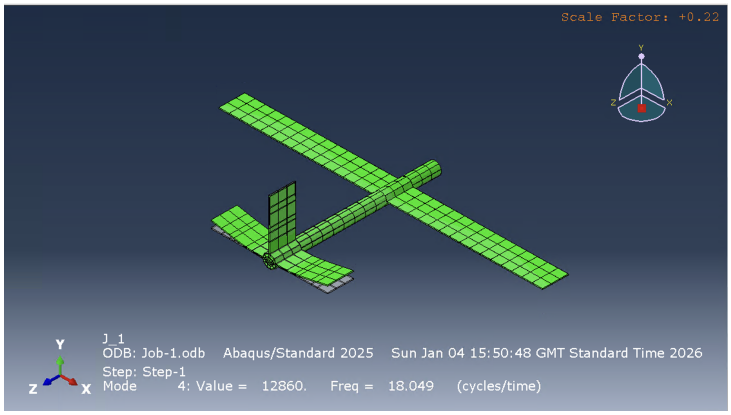


Figure 7: Abaqus eigenmode 4: at 18.0 Hz showing coupled wing–fuselage motion.

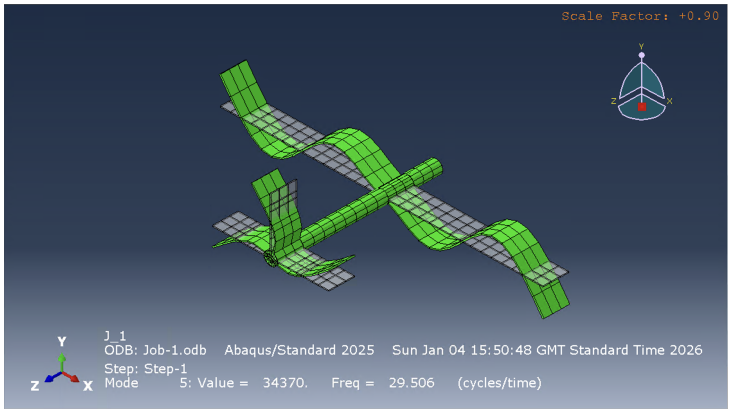


Figure 8: Abaqus eigenmode 5: at 29.5 Hz corresponding closely to the experimentally observed resonance.

6 Comparison and Discussion:

The experimentally identified eigenfrequencies were found to lie in the range of 28 30 Hz, which compares well with the fifth eigenmode predicted by the Abaqus simulation at approximately 29.5 Hz. The corresponding mode shapes also show good qualitative agreement, with similar bending behaviour observed in the wings and fuselage. Lower frequency numerical modes in the range of 12 18 Hz were not as clearly identified during the experimental testing. This is likely due to limited excitation strength, measurement resolution, and overlap between neighbouring modes. In addition, the experimentally reconstructed mode shapes display minor asymmetry and noise, which can be attributed to measurement uncertainty, imperfect experimental conditions, and structural damping. Overall, despite these differences, the model successfully captures the dominant dynamic behaviour of the structure and provides a reasonable representation of the observed experimented eigenmodes.

7 Conclusions:

This laboratory exercise demonstrated the use of both experimental testing and numerical simulation to identify eigenfrequencies and mode shapes of an aircraft structure. Experimental measurements provided direct insight into the dynamic behaviour of the structure, while Abaqus CAE was used to predict the corresponding eigenfrequencies and eigenmodes using a finite element approach. Good agreement was observed between the experimental and numerical results for the dominant resonance near 30 Hz. Differences at lower frequencies, which highlight the practical limitations of experimental measurements and the simplifying assumptions used in the numerical model. Overall, the results show that combining experimental testing with finite element analysis is an effective approach for analysing structural vibration in aerospace applications.